

HOSPITAL MANAGEMENT SYSTEM

Final Project Report

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|Modern Application Development - I |

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Course: Modern Application Development - I

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I am a college student focusing on programming and data science. This project was an excellent practical learning opportunity. I encountered several challenging points, most notably with the logic for multi-user authentication. Also getting the availability logic right took time, but was satisfying. Styling has never been my cup of tea but I have taken time to make it look presentable. Working through and resolving those complex issues was a valuable part of the process and provided a great sense of accomplishment.

Description

In this project, I built a **Hospital Management System (HMS)** using HTML, CSS, Bootstrap, Flask, and SQLAlchemy. The application supports three distinct user roles:

Admin, Doctor, and Patient.

A core feature is the robust login/signup system where each user role has a separate dashboard and permissions. Patients can create accounts, browse departments, view doctor profiles, and book appointments based on the doctor's real-time availability. Doctors can log in to manage their schedules, set their weekly availability using a toggle interface, and update patient history after an appointment. The Admin serves as the superuser, with a dashboard to manage all doctor and patient accounts, including creating, editing, and deleting them.

AI/LLM Declaration (23%)

Styling/UI – 8%

HTML + Jinja2 – 15%

Technologies Used

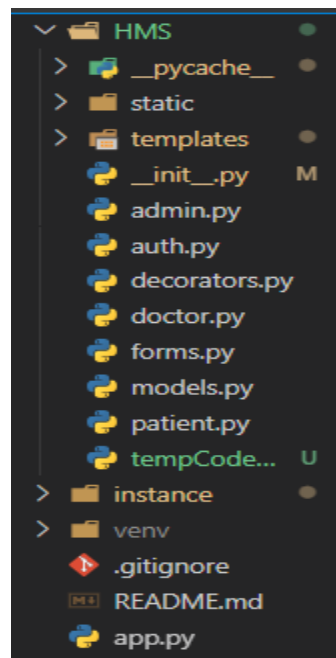
1. **Flask:** Used for building the core web application and routes.
2. **Flask-SQLAlchemy:** Extension used to handle the database (models, schema, and sessions).
3. **Flask-Login:** Used for user session management, handling `current_user`, and route protection.

4. **Flask-WTF:** Used to create, validate, and render all forms (Login, Profile Edit, Availability, Booking).
5. **Flask-Bcrypt:** Used to securely hash and check user passwords.
6. **Flask-Migrate:** Used to manage database schema changes and upgrades.
7. **Bootstrap 5:** Frontend framework used for all styling, layout (grid, cards, buttons), and components.
8. **Datetime:** This Python module was used extensively to manage dates and times for appointments and availability.
9. **OS (Operating System):** This module was used to check for the database file's existence using `os.path.exists()`. This check, placed in our `__init__.py`, prevents the app from re-creating the database and default admin on every server restart.

ARCHITECTURE

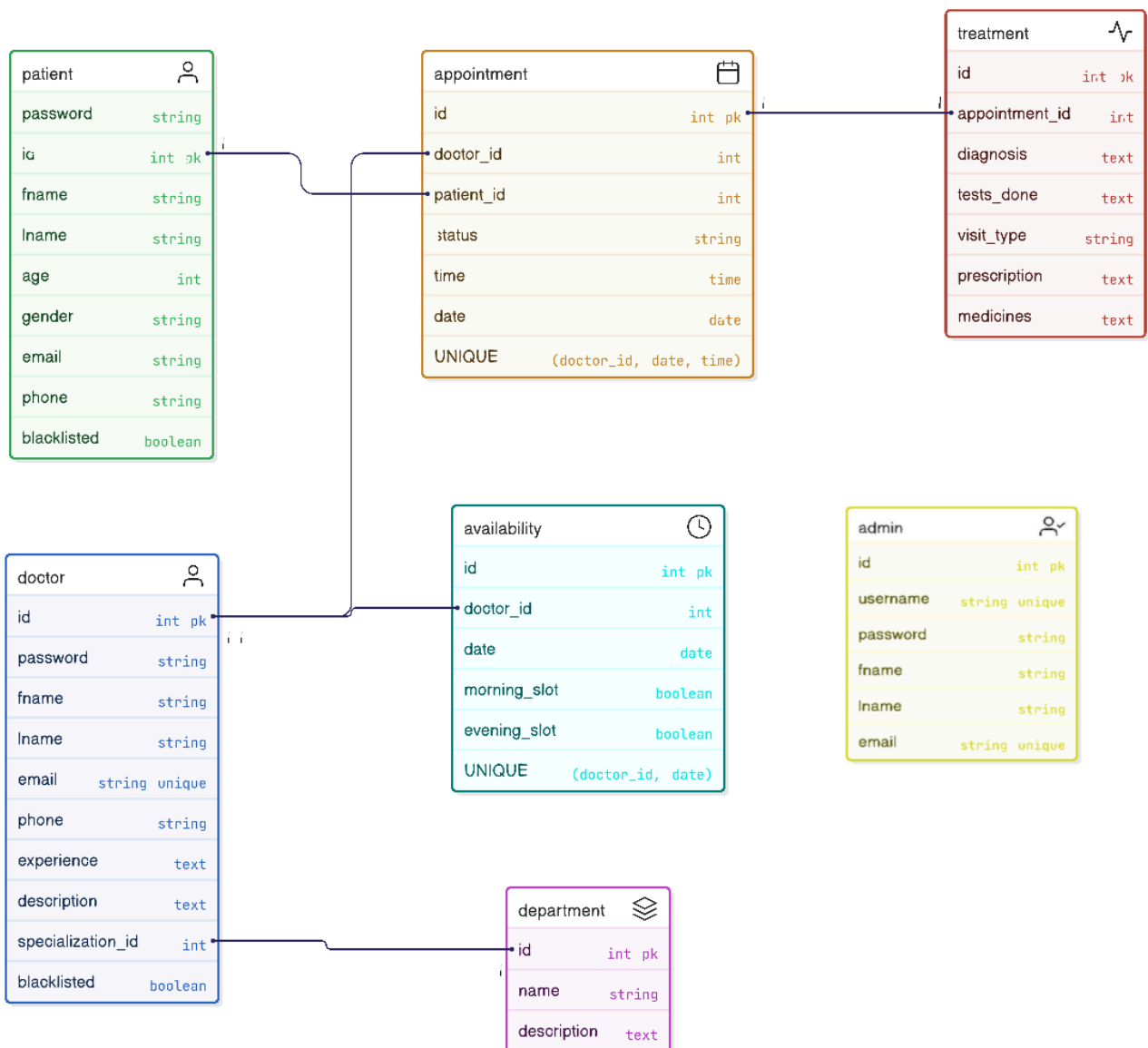
Here, The Main.py file contains the main code to run the webapp.LMS(Library Management System) is a module which contains all the files for the app. The description of files inside LMS are as follows: -

- **Auth.py** – This file contains Authorization logic like login, signup for patient/doctor
- **Admin.py** – This contain all functionalities related to Admin
- **Doctor.py** – This contain all functionalities related to Doctor
- **Patient.py** – This contains all functionalities related to Patient
- **Models.py** – This contains DB Classes
- **Forms.py** – This contains all the form used in application
- **Decorators.py** – This contains decorator like, `@admin_required`, `@doctor_required` etc.
- **Templates** – This contains all HTML templates for Admin, Doctor and Patient
- **Static** – This contains CSS and image files



DB Schema Design

The database schema consists of tables for patient, doctor, admin, appointments, treatment, availability, and department. Relationships are established between these tables to track user activities such as upcoming appointments, patient history, doctor's availability, and treatments of a patient. Each table contains relevant fields, ensuring comprehensive data management within the Hospital Management system. There exists different relationships between tables, as one-to-one, one-to-many, and multiple. And then there is the Admin table which doesn't share any relationship with any table.



- VIDEO LINK  project_video.mp4

A video demonstration of my project is available here.

<https://drive.google.com/file/d/1Au1mpVe6lg1OvY7eh3dAWxHz2eVhUK-/view?usp=sharing>

THANK YOU

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